Recap: Lecture 28

In Lecture 28, we

- Introduced the Fourier transform and inverse FT
- Connected the FT with Fourier series
- Gave a theorem for existence of the FT

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$$\mathscr{F}\{f\} = \int_{-\infty}^{\infty} f(x)\,\mathrm{e}^{-\mathrm{i}\,k\,x}\,\mathrm{d}x = \widehat{f}(k)$$

$$\mathscr{F}^{-1}\{f\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(k)\,\mathrm{e}^{+\mathrm{i}\,k\,x}\,\mathrm{d}k$$

Lecture 29

In Lecture 29, we will:

1. Introduce a FT table for simpler evaluation of FTs and inverse FTs
2. Introduce convolution and shift theorems for FT
3. Learn rules to help apply FTs to ODEs

Table of Fourier transforms

$f(x)$	$\widehat{f}(k)$
$\delta_a(x) \quad (a \in \mathbb{R})$	$\mathrm{e}^{-\mathrm{i}ak}$
$\mathrm{e}^{-a\lvert x\rvert} \quad (a > 0)$	$\dfrac{2a}{a^2+k^2}$
$H_{-a}(x) - H_a(x) \quad (a > 0)$	$2a \operatorname{sinc} ak$
$\mathrm{e}^{-ax^2} \quad (a > 0)$	$\sqrt{\dfrac{\pi}{a}}\mathrm{e}^{-\frac{k^2}{4a}}$

$f(x)$	$\widehat{f}(k)$
$\mathrm{e}^{\mathrm{i}ax} \quad (a \in \mathbb{R})$	$2\pi\delta_a(k)$
$\dfrac{1}{a^2+x^2} \quad (a > 0)$	$\dfrac{\pi}{a}\mathrm{e}^{-a\lvert k\rvert}$
$\operatorname{sinc}(ax) \quad (a > 0)$	$\dfrac{\pi}{a}\left(H_{-a}(k) - H_a(k)\right)$
$\operatorname{sech}(ax) \quad (a \neq 0)$	$\dfrac{\pi}{a}\operatorname{sech}\left(\dfrac{\pi k}{2a}\right)$

$$^*\operatorname{sinc}(y) = \begin{cases} \frac{\sin y}{y}, & y \neq 0 \\ 1, & y = 0 \end{cases}$$

Exercise 6.4: Find the inverse Fourier transform of $\widehat{f}(k) = \sin(2k)$

$$\sin(x) = \frac{1}{2i}\left(e^{ix} - e^{-ix}\right) \Rightarrow \sin(2k) = \frac{1}{2i}\left(e^{2ik} - e^{-2ik}\right)$$

$$F^{-1}(\sin(2k)) = F^{-1}\left\{\frac{1}{2i}\left(e^{2ik} - e^{-2ik}\right)\right\}$$

$$= \frac{1}{2i}\left\{F^{-1}(e^{2ik}) - F^{-1}(e^{-2ik})\right\}$$

use FT table:

$$= \frac{1}{2i}\left(\delta_{-2}(x) - \delta_2(x)\right)$$

$$= \frac{1}{2i}\left(\delta(x+2) - \delta(x-2)\right)$$

Recap: Convolutions for Laplace transforms

- Given two functions, f and g, defined on $[0, \infty)$, their convolution, $f * g$ is defined

$$(f * g)(t) = \int_0^t f(t - \tau) g(\tau) \, \mathrm{d}\tau$$

- Laplace transforms convert convolutions into multiplication, i.e.,

$$\mathscr{L}\{f * g\} = F(s)G(s)$$

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Convolutions for Fourier transforms

- We define the convolution of $f(x)$ with $g(x)$ for $x \in \mathbb{R}$ as

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- Then the Fourier transform of the convolution is

$$\mathscr{F}\{f * g\} = \widehat{f}(k) \, \widehat{g}(k)$$

Exercise 6.5: Calculate the convolution of

Table

$$\mathcal{F}\left(\frac{1}{a^2+x^2}\right) = \frac{\pi}{a} e^{-a|k|}$$

$$f(x) = \frac{1}{x^2+4} \quad \text{with} \quad g(x) = \frac{1}{x^2+9}$$

case $a = 2$

$$\hat{f}(k) = \frac{\pi}{2} e^{-2|k|}$$

case $a = 3$

$$\hat{g}(k) = \frac{\pi}{3} e^{-3|k|}$$

$$\mathcal{F}\{f*g\} = \frac{\pi}{2} e^{-2|k|} \cdot \frac{\pi}{3} e^{-3|k|} = \frac{\pi^2}{6} \cancel{e^{-2|k|-3|k|}} \; e^{-5|k|}$$

$$f*g = \mathcal{F}^{-1}\left\{\frac{\pi^2}{6} e^{-5|k|}\right\} = \frac{5}{6}\pi \, \mathcal{F}^{-1}\left\{\frac{\pi}{5} e^{-5|k|}\right\} = \frac{5\pi}{6} \frac{1}{5^2+x^2}$$

Recap: Laplace transform shift theorems

Theorem (t-shift)

If $F(s)$ exists for $\mathrm{Re}(s) > \alpha$, and $c \geqslant 0$, then

$$\mathscr{L}\Big\{f(t-c)H(t-c)\Big\} = \mathrm{e}^{-cs}F(s), \quad \mathrm{Re}(s) > \alpha$$

Theorem (s-shift)

If $\mathscr{L}\{f\}$ exists for $\mathrm{Re}(s) > \alpha$, then

$$\mathscr{L}\{\mathrm{e}^{\omega t}f(t)\} = F(s-\omega), \qquad \mathrm{Re}(s) > \alpha + \omega$$

Fourier transform shift theorems

1. $\mathscr{F}\{f(x-y)\} = \mathrm{e}^{-iky}\,\widehat{f}(k) \qquad (y \in \mathbb{R})$
2. $\mathscr{F}\{\mathrm{e}^{iqx} f\} = \widehat{f}(k-q) \qquad (q \in \mathbb{R})$

Fourier transform shift theorems

$$\mathcal{F}\{e^{-a|x|}\} = 2a/(a^2+k^2)$$

$$\mathcal{F}^{-1}\left\{\frac{\pi}{a^2}\operatorname{sech}\left(\frac{\pi}{2a}k\right)\right\} = \operatorname{sech}(ax)$$

1. $\mathscr{F}\{f(x-y)\} = e^{-iky}\,\widehat{f}(k) \qquad (y \in \mathbb{R})$

2. $\mathscr{F}\{e^{iqx} f\} = \widehat{f}(k-q) \qquad (q \in \mathbb{R})$

$$e^{iqx} f(x) = \mathcal{F}^{-1}\{\hat{f}(k-q)\}$$

$$\mathcal{F}^{-1}\left\{\pi \operatorname{sech}\left(\frac{\pi k}{2}\right)\right\} = \operatorname{sech}(x)$$

Exercise 6.6: Evaluate the following

$\mathscr{F}\{e^{-a|x-2|}\}$

$$f(x) = e^{-a|x|} \qquad \mathcal{F}\{f(x-2)\} = e^{-2ik}\hat{f}(k) = \frac{2ae^{-2ik}}{a^2+k^2}$$

$$\mathscr{F}^{-1}\left\{\operatorname{sech}\frac{\pi(k-5)}{2}\right\} = e^{iqx} f(x) = e^{5ix} f(x) = \frac{e^{5ix}\operatorname{sech}(x)}{\pi}$$

we have $\hat{f}(k) = \operatorname{sech}\frac{\pi k}{2}$

Application to ODEs

Just like Laplace transforms, Fourier transforms turns calculus into algebra!

1. Differentiation wrt x turns into multiplication by k

$$\mathscr{F}\left(f^{(n)}\right) = (ik)^n \widehat{f}$$

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2. Multiplication by x turns into differentiation w.r.t. k

$$\mathscr{F}(x^n f) = i^n \frac{\mathrm{d}^n \widehat{f}}{\mathrm{d}k^n}$$

Exercise 6.7: Show that $\mathscr{F}(f') = ik\mathscr{F}(f)$

$$F(f') = \int_{-\infty}^{\infty} f'(x) e^{-ikx} dx = \lim_{a \to \infty} \int_{-a}^{a} f'(x) e^{-ikx} dx$$

$$= \lim_{a \to \infty} \left\{ \cancelto{0}{f(x) e^{-ikx} \Big|_{-a}^{a}} - \int_{-a}^{a} f(x) \cdot -ik e^{-ikx} dx \right\}$$

$$= \lim_{a \to \infty} ik \int_{-a}^{a} f(x) e^{-ikx} dx$$

$$= ik F\{f\}$$

Exercise 6.8: Show that $\mathscr{F}(x\,f) = i\frac{\mathrm{d}\hat{f}}{\mathrm{d}k}$

$$\frac{d}{dk}\hat{f}(k) = \frac{d}{dk}\int_{-\infty}^{\infty} f(x)e^{-ikx}\,dx$$

$$= \int_{-\infty}^{\infty} f(x)\underbrace{\left(\frac{d}{dk}e^{-ikx}\right)}_{-ix\,e^{-ikx}}dx$$

$$= -i\int_{-\infty}^{\infty} x f(x) e^{-ikx}\,dx = -i\,F\{xf\}$$

$$\therefore \cancel{i\cdot -i}\,F\{xf\} = i\frac{df}{dk}$$

Exercise 6.9: Find the solution to the ODE

$$y''(x) - y(x) = \delta(x)$$

subject to the condition that the Fourier transform of $y(x)$ exists.

Apply FT to both sides of ODE:

RHS: $\underbrace{F\{\delta(x)\}}_{\delta_0(x) \to \text{ in table}} = e^{-i \cdot 0 \cdot k} = e^0 = 1$

LHS: $F\{y'' - y\} = F\{y''\} - F\{y\} = -k^2 \hat{y} - \hat{y}$

$= -(1+k^2)\hat{y}$

$\therefore \quad -(1+k^2)\hat{y} = 1 \quad \Rightarrow \quad \hat{y} = \dfrac{-1}{1+k^2}$

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subject to the condition that the Fourier transform of $y(x)$ exists.

$$\therefore \quad F^{-1}\left\{ \frac{-1}{1+k^2} \right\} = \frac{-1}{2} F\left\{ \frac{2}{1+k^2} \right\}$$

$$= \frac{-1}{2} e^{-|x|} = y(x)$$